

Naive Bayes in Classification

Naive Bayes is a **supervised machine learning classification technique** based on **Bayes' Theorem**.

It is used to classify data into different classes by calculating the **probability of each class** and selecting the class with the highest probability.

The word “**Naive**” means that the algorithm assumes that the features are **independent of each other**.

Naive Bayes predicts the class of a new data item by using the probability of its features and selecting the most probable class.

Example: Email Classification

Suppose an email contains the words:

“Win free money”

We want to classify it as:

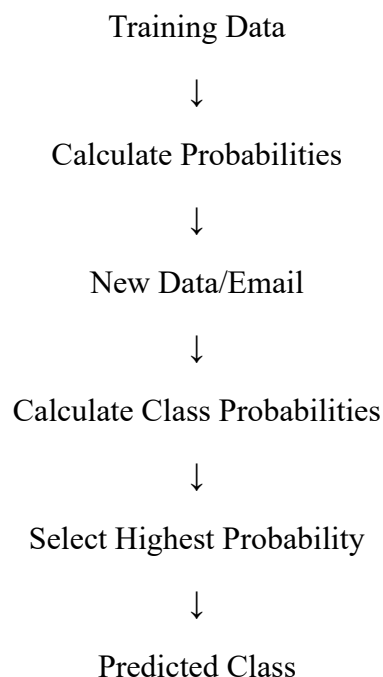
- **Spam**
- **Not Spam**

Naive Bayes checks the probability of these words occurring in spam and non-spam emails.

If: **Probability of Spam > Probability of Not Spam**

then the email is classified as: **Spam**

Working



Bayes' Theorem



Where:

- $P(A|B)$ = Probability of A when B is known
- $P(B|A)$ = Probability of B when A is known
- $P(A)$ = Probability of A
- $P(B)$ = Probability of B

Applications of Naive Bayes

1. Spam email detection
2. Text classification
3. Sentiment analysis
4. Document classification
5. News classification
6. Medical diagnosis

Advantages

- Simple and easy to implement.
- Fast in training and prediction.
- Works well with large datasets.
- Particularly useful for text classification.
- Requires relatively less training data compared with some other methods.

Naive Bayes is a supervised classification algorithm based on Bayes' Theorem that assumes the features are independent and classifies a data item according to the class having the highest probability.

Naive Bayes – Numerical Example

Suppose we want to classify an email as **Spam** or **Not Spam**.

We have the following training data:

Email	Free	Class
1	Yes	Spam
2	Yes	Spam
3	No	Not Spam
4	No	Not Spam
5	Yes	Spam

Now, classify a new email:

Free = Yes

Step 1: Calculate Prior Probability

Total emails = 5

Spam emails = 3

Not Spam emails = 2

$P(\text{Spam}) = 3 / 5 = 0.6$

$P(\text{Not Spam}) = 2 / 5 = 0.4$

Step 2: Calculate Conditional Probability

Among 3 Spam emails, all 3 contain “Free = Yes”.

$P(\text{Free}|\text{Spam}) = 3 / 3 = 1$

Among 2 Not Spam emails, none contains “Free = Yes”.

$P(\text{Free} | \text{Not Spam}) = 0 / 2 = 0$

Step 3: Apply Bayes' Formula

Bayes' theorem:

$P(\text{Class} | \text{Feature}) = (P(\text{Feature} | \text{Class}) * P(\text{Class})) / P(\text{Feature})$

For Spam:

$$P(\text{Spam} | \text{Free}) = (P(\text{Free} | \text{Spam}) * P(\text{Spam})) / P(\text{Free})$$
$$= (1 * 0.6) / P(\text{Free})$$

For Not Spam:

$$P(\text{Not Spam} | \text{Free}) = (0 * 0.4) / P(\text{Free}) = 0$$

Since the probability of **Spam** is higher, the new email is classified as:

Spam

Rule: - Calculate probability for each class → Compare probabilities → Choose the class with the highest probability.

Clustering Technique – K-Means

K-Means is an **unsupervised machine learning clustering technique** used to divide a dataset into **K groups (clusters)** based on the similarity of data points.

In K-Means, each data point is assigned to the cluster whose **centroid (center)** is nearest to it.

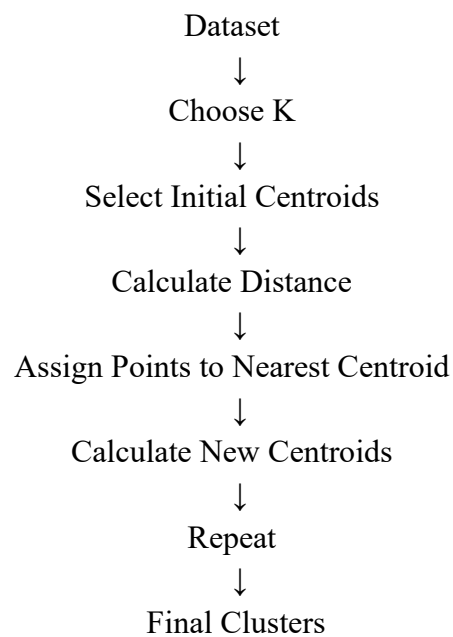
Important Terms

- **K** → Number of clusters required.
- **Centroid** → Center point of a cluster.
- **Cluster** → Group of similar data points.
- **Distance** → Used to determine which centroid is closest to a data point.

Working of K-Means

1. Choose the number of clusters **K**.
2. Select **K initial centroids**.
3. Calculate the distance of each data point from each centroid.
4. Assign each data point to the **nearest centroid**.
5. Recalculate the centroid of each cluster.
6. Repeat the assignment and centroid calculation.
7. Stop when the centroids become stable or there is no significant change.

Flow



Distance Formula

Usually, **Euclidean Distance** is used:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

where:

- d = Distance between two points
- x_1, y_1 = Coordinates of first point
- x_2, y_2 = Coordinates of second point

Example

Suppose we have students based on their **study hours and marks**:

Student	Study Hours	Marks
A	1	30
B	2	35
C	3	40
D	8	80
E	9	85
F	10	90

Let:

$K=2$

K-Means can form two clusters:

Cluster 1: A, B, C

Students with lower study hours and marks.

Cluster 2: D, E, F

Students with higher study hours and marks.

Thus, K-Means automatically groups similar students into different clusters.

Applications

1. Customer segmentation
2. Image segmentation
3. Document clustering
4. Market analysis
5. Student performance grouping
6. Recommendation systems
7. Pattern recognition
8. Anomaly detection

Advantages

- Simple and easy to implement.
- Fast for large datasets.
- Easy to understand.
- Works well when clusters are clearly separated.
- Computationally efficient.

Disadvantages

- Value of **K** must be chosen in advance.
- Sensitive to initial centroid selection.
- Sensitive to outliers.
- Does not work well with irregularly shaped clusters.
- Different initial centroids may produce different results.

K-Means is an unsupervised clustering algorithm that divides data into K clusters by assigning each data point to the nearest centroid and repeatedly updating the centroids until the clusters become stable.

2-Dimensional K-Means Numerical Example

Question

Consider the following 2-dimensional data points:

Point	X	Y
A	1	1
B	2	2
C	3	3
D	8	8
E	9	9
F	10	10

Take:

K=2

Let the initial centroids be:

$$C_1 = (1, 1)$$

$$C_2 = (8, 8)$$



Step 1: Calculate Euclidean Distance

For a 2-dimensional point:

$$d = \sqrt{(x - x_c)^2 + (y - y_c)^2}$$

where (x, y) is the data point and (x_c, y_c) is the centroid.

Step 2: Calculate Distance of Each Point

Point A = (1,1)

Distance from $C_1 = (1, 1)$:

$$\begin{aligned}d(A, C_1) &= \sqrt{(1-1)^2 + (1-1)^2} \\&= 0\end{aligned}$$

Distance from $C_2 = (8, 8)$:

$$\begin{aligned}d(A, C_2) &= \sqrt{(1-8)^2 + (1-8)^2} \\&= \sqrt{49 + 49} = \sqrt{98} = 9.90\end{aligned}$$

Therefore, **A → Cluster 1**

Point B = (2,2)

$$\begin{aligned}d(B, C_1) &= \sqrt{(2-1)^2 + (2-1)^2} \\&= \sqrt{2} = 1.41\end{aligned}$$

$$\begin{aligned}d(B, C_2) &= \sqrt{(2-8)^2 + (2-8)^2} \\&= \sqrt{72} = 8.49\end{aligned}$$

Therefore, **B → Cluster 1**

Point C = (3,3)

$$\begin{aligned}d(C, C_1) &= \sqrt{(3-1)^2 + (3-1)^2} \\&= \sqrt{8} = 2.83\end{aligned}$$

$$\begin{aligned}d(C, C_2) &= \sqrt{(3-8)^2 + (3-8)^2} \\&= \sqrt{50} = 7.07\end{aligned}$$

Therefore, **C → Cluster 1**

Point D = (8,8)

$$\begin{aligned}d(D, C_1) &= \sqrt{(8-1)^2 + (8-1)^2} \\&= \sqrt{98} = 9.90\end{aligned}$$

$$d(D, C_2) = 0$$

Therefore, **D → Cluster 2**

Point E = (9,9)

$$\begin{aligned}d(E, C_1) &= \sqrt{(9-1)^2 + (9-1)^2} \\&= \sqrt{128} = 11.31\end{aligned}$$

$$\begin{aligned}d(E, C_2) &= \sqrt{(9-8)^2 + (9-8)^2} \\&= \sqrt{2} = 1.41\end{aligned}$$

Therefore, **E → Cluster 2**



Point F = (10,10)

$$d(F, C_1) = \sqrt{(10 - 1)^2 + (10 - 1)^2}$$

$$= \sqrt{162} = 12.73$$

$$d(F, C_2) = \sqrt{(10 - 8)^2 + (10 - 8)^2}$$

$$= \sqrt{8} = 2.83$$

Therefore, F → Cluster 2

Step 3: Form Clusters

Cluster	Points
Cluster 1	A, B, C
Cluster 2	D, E, F

Step 4: Calculate New Centroids

New Centroid of Cluster 1

Points:

$$(1, 1), (2, 2), (3, 3)$$

Formula:

$$C_1 = \left(\frac{\sum X}{n}, \frac{\sum Y}{n} \right)$$

$$C_1 = \left(\frac{1 + 2 + 3}{3}, \frac{1 + 2 + 3}{3} \right)$$

$$C_1 = (2, 2)$$

New Centroid of Cluster 2

Points:

$$(8, 8), (9, 9), (10, 10)$$

$$C_2 = \left(\frac{8 + 9 + 10}{3}, \frac{8 + 9 + 10}{3} \right)$$

$$C_2 = (9, 9)$$

Final Result

$$C_1 = (2, 2)$$

$$C_2 = (9, 9)$$

Therefore:

Cluster 1 $\rightarrow$ A (1,1), B (2,2), C (3,3)

Cluster 2 $\rightarrow$ D (8,8), E (9,9), F (10,10)

What Happens After Calculating New Centroids?

$$C_1 = (2, 2)$$

$$C_2 = (9, 9)$$

Now we **repeat the same process** using these **new centroids**.

1. Again Calculate Distances

For every point, calculate its distance from the **new** C_1 and **new** C_2 .

For example, for point C = (3,3):

Distance from $C_1 = (2, 2)$:

$$\begin{aligned} d(C, C_1) &= \sqrt{(3-2)^2 + (3-2)^2} \\ &= \sqrt{2} = 1.41 \end{aligned}$$

Distance from $C_2 = (9, 9)$:

$$\begin{aligned} d(C, C_2) &= \sqrt{(3-9)^2 + (3-9)^2} \\ &= \sqrt{72} = 8.49 \end{aligned}$$

So C remains in **Cluster 1**.

2. Do This for All Points

Point	Distance from C_1 (2,2)	Distance from C_2 (9,9)	Cluster
A (1,1)	1.41	11.31	C_1
B (2,2)	0	9.90	C_1
C (3,3)	1.41	8.49	C_1
D (8,8)	8.49	1.41	C_2
E (9,9)	9.90	0	C_2
F (10,10)	11.31	1.41	C_2

So, the clusters are **still**:

Cluster1 = {A, B, C}

Cluster2 = {D, E, F}

3. Calculate Centroids Again

Since the clusters have not changed:

$$C_1 = \left(\frac{1+2+3}{3}, \frac{1+2+3}{3} \right)$$
$$C_1 = (2, 2)$$

And:

$$C_2 = \left(\frac{8+9+10}{3}, \frac{8+9+10}{3} \right)$$
$$C_2 = (9, 9)$$

The centroids are **exactly the same as before**.

4. Stop the Algorithm

Now we compare:

Old centroids:

$$C_1 = (2, 2), \quad C_2 = (9, 9)$$

New centroids:

$$C_1 = (2, 2), \quad C_2 = (9, 9)$$

There is **no change**.

Therefore, K-Means stops.

Final Clusters

$$\boxed{\text{Cluster 1} = \{A, B, C\}}$$

$$\boxed{\text{Cluster 2} = \{D, E, F\}}$$

Complete K-Means Process

